OpenAI Platform

Agents

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Learn how to build agents with the OpenAI API.

Agents represent **systems that intelligently accomplish tasks**, ranging from executing simple workflows to pursuing complex, open-ended objectives.

OpenAI provides a **rich set of composable primitives that enable you to build agents**. This guide walks through those primitives, and how they come together to form a robust agentic platform.

Overview

Building agents involves assembling components across several domains—such as **models, tools, knowledge and memory, audio and speech, guardrails, and orchestration**—and OpenAI provides composable primitives for each.

DOMAIN	DESCRIPTION	OPENAI PRIMITIVES
Models	Core intelligence capable of reasoning, making decisions, and processing different	o1, o3-mini, GPT-4.5, GPT-4o, GPT-4o-mini

	modalities.	
Tools	Interface to the world, interact with environment, function calling, built-in tools, etc.	Function calling, Web search, File search, Computer use
Knowledge and memory	Augment agents with external and persistent knowledge.	Vector stores, File search, Embeddings
Audio and speech	Create agents that can understand audio and respond back in natural language.	Audio generation, realtime, Audio agents
Guardrails	Prevent irrelevant, harmful, or undesirable behavior.	Moderation, Instruction hierarchy
Orchestration	Develop, deploy, monitor, and improve agents.	Agents SDK, Tracing, Evaluations, Fine-tuning
Voice agents	Create agents that can understand audio and respond back in natural language.	Realtime API, Voice support in the Agents SDK

Models

MODEL	AGENTIC STRENGTHS

o3 and o4-mini	Best for long-term planning, hard tasks, and reasoning.
GPT-4.1	Best for agentic execution.
GPT-4.1-mini	Good balance of agentic capability and latency.
GPT-4.1-nano	Best for low-latency.

Large language models (LLMs) are at the core of many agentic systems, responsible for making decisions and interacting with the world. OpenAI's models support a wide range of capabilities:

- **High intelligence:** Capable of reasoning and planning to tackle the most difficult tasks.
- **Tools:** Call your functions and leverage OpenAI's built-in tools.
- **Multimodality:** Natively understand text, images, audio, code, and documents.
- **Low-latency:** Support for real-time audio conversations and smaller, faster models.

For detailed model comparisons, visit the models page.

Tools

Tools enable agents to interact with the world. OpenAI supports **function calling** to connect with your code, and **built-in tools** for common tasks like web searches and data retrieval.

TOOL	DESCRIPTION

Function calling	Interact with developer-defined code.
Web search	Fetch up-to-date information from the web.
File search	Perform semantic search across your documents.
Computer use	Understand and control a computer or browser.

Knowledge and memory

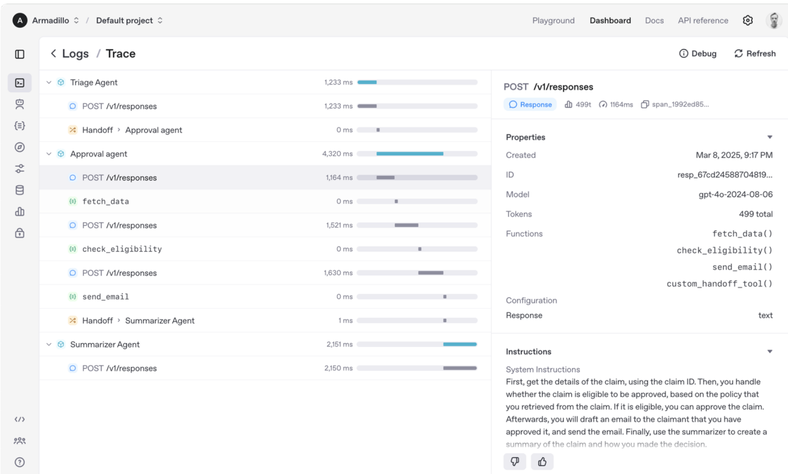
Knowledge and memory help agents store, retrieve, and utilize information beyond their initial training data. **Vector stores** enable agents to search your documents semantically and retrieve relevant information at runtime. Meanwhile, **embeddings** represent data efficiently for quick retrieval, powering dynamic knowledge solutions and long-term agent memory. You can integrate your data using OpenAI's vector stores and Embeddings API.

Guardrails

Guardrails ensure your agents behave safely, consistently, and within your intended boundaries—critical for production deployments. Use OpenAI's free Moderation API to automatically filter unsafe content. Further control your agent's behavior by leveraging the instruction hierarchy, which prioritizes developer-defined prompts and mitigates unwanted agent behaviors.

Orchestration

Building agents is a process. OpenAI provides tools to effectively build, deploy, monitor, evaluate, and improve agentic systems.



PHASE	DESCRIPTION	OPENAI PRIMITIV
Build and deploy	Rapidly build agents, enforce guardrails, and handle conversational flows using the Agents SDK.	Agents SDK
Monitor	Observe agent behavior in real-time, debug issues, and gain insights through tracing.	Tracing
Evaluate and improve	Measure agent	Evaluations Fine-tuning

performance, identify areas for improvement, and refine your agents.

Get started

Get started by installing the OpenAI Agents SDK for Python via:

```
pip install openai-agents
```

Explore the repository and documentation for more details.